Publication Decision from *Journal of High School Science*

"Emerging success of flexible strategies in increasingly noisy Iterated Prisoner's Dilemma"

Decision made on July 13, 2026

Editorial board's determination

Revise and resubmit

Comments from the editor

Dear author,
Please address all the comments of the reviewer and revise your manuscript accordingly.
When you resubmit, **please also submit separately a word file that lists verbatim each comment by the reviewer and then describe how and where in the revised manuscript you have addressed that particular comment. Do this for all comments in one separate file.**
Please note that if the the concerns/comments/questions of the reviewer are addressed inadequately, incompletely or insufficient address is paid to detail, your manuscript may be rejected. Please therefore thoroughly review your responses before resubmission.
We look forward to reviewing your revised manuscript.
Best,
Shireesh Apte

Reviewer 1

Rating scale questions

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The paper makes a significant contribution to scholarship.		✓			

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The literature review was thorough given the objectives and content of the article.			✓		

Open response questions

Comments to author

I would recommend major revision before this manuscript is suitable for publication. The study addresses a worthwhile question—the behavior of classical Iterated Prisoner's Dilemma (IPD) strategies under increasing implementation noise—and the computational framework is clearly presented. However, the manuscript substantially overstates the novelty of its central conclusion while providing insufficient evidence that the proposed concept of "flexibility" represents a genuinely new strategic principle. The literature on noisy IPD has, for several decades, focused on precisely this problem: identifying strategic properties that permit robust cooperation in the presence of implementation errors. Concepts such as generosity, contrition, probabilistic forgiveness, stochastic decision-making, and robustness to noise have already been investigated extensively. Although the manuscript cites this prior work, it does not adequately position its contribution relative to it, giving the impression that "flexibility" is a newly discovered property rather than a reinterpretation or synthesis of existing concepts. The authors should explicitly explain how flexibility differs from, extends, or unifies previously recognized mechanisms instead of presenting it as an entirely novel discovery.

A second major concern is that the proposed concept of flexibility lacks a rigorous operational definition. Throughout the Discussion, flexibility is variously described as (1) probabilistic decision-making, (2) reliance on short memory, and (3) reduced sensitivity to isolated noisy defections. These are distinct algorithmic properties that need not coincide. A probabilistic strategy may have long memory, while a deterministic strategy may have extremely short memory, yet both could potentially satisfy parts of the proposed definition. Consequently, the manuscript currently defines flexibility descriptively rather than mechanistically. The authors should formulate a single, precise definition that can be objectively evaluated across strategies and distinguish it clearly from existing concepts such as forgiveness, stochasticity, memory length, or adaptive updating.

The evidence supporting flexibility as a general principle is also limited. The primary observation is that Downing becomes the highest-performing strategy as noise increases, whereas several historically successful strategies decline in performance. While this is an interesting empirical result, one successful strategy is insufficient to establish a universal design principle. The manuscript effectively infers the existence of flexibility after observing the tournament outcomes, then retrospectively classifies successful strategies as flexible and unsuccessful strategies as rigid. This post hoc reasoning weakens the scientific argument. A stronger methodology would define flexibility before any simulations are performed, classify all strategies according to objective criteria, and then statistically evaluate whether flexibility predicts tournament performance. Without this prospective approach, it remains unclear whether flexibility genuinely explains the observed rankings or simply describes them after the fact.

Relatedly, flexibility is never quantified despite the manuscript proposing an intuitive perturbation test in which a single cooperation is replaced by a defection to observe the long-term behavioral consequences. This idea has considerable potential but remains entirely qualitative. The manuscript would be substantially strengthened by introducing a quantitative flexibility metric—for example, the expected recovery time following a perturbation, the persistence length of retaliation chains, a sensitivity coefficient measuring behavioral change after a noisy move, or another formally defined robustness measure. Such a metric would permit statistical analyses, including regression between flexibility and tournament performance, thereby transforming flexibility from a descriptive label into a measurable scientific variable.

Conceptually, the manuscript may also benefit from reframing its interpretation. The behavior described as flexibility closely resembles robustness to noisy observations, a problem that has been extensively studied in statistical decision theory, signal processing, and control theory. Noise randomly corrupts observations, and successful strategies effectively filter transient errors rather than reacting aggressively to every isolated event. Rigid strategies behave analogously to high-gain controllers that amplify measurement noise into prolonged retaliation cycles, whereas flexible strategies resemble low-pass filters that integrate information over multiple observations before responding. Presenting the findings within this broader framework of robustness to observation noise would place the results within an established theoretical context and avoid implying that an entirely new strategic principle has been discovered.

The experimental design could also be strengthened. Only the fifteen strategies from Axelrod's original tournament were evaluated. Although these strategies are historically important, they represent only a small subset of the extensive modern IPD literature. Numerous later strategies—including Generous Tit-for-Tat, Contrite Tit-for-Tat, Win-Stay Lose-Shift (Pavlov), adaptive memory-based strategies, and reinforcement-learning approaches—were specifically developed to address noisy environments. Demonstrating that flexibility predicts performance across this broader strategy space would provide substantially stronger support for the proposed hypothesis and help establish whether the observed phenomenon is general rather than an artifact of the historical strategy set selected for analysis.

The statistical treatment of the results is currently insufficient to support the strength of the manuscript's conclusions. Tournament scores are averaged across ten repetitions, but measures of uncertainty such as standard deviations, confidence intervals, or hypothesis tests are not reported. Similarly, no statistical assessment is performed to determine whether changes in rankings or score differences are significant across noise levels. Because stochastic simulations inherently exhibit variability, quantitative statistical analysis is necessary before concluding that one strategic characteristic consistently outperforms another.

Finally, several aspects of the interpretation should be moderated. The recreated tournament does not reproduce Axelrod's original ranking, with Nydegger outperforming Tit for Tat under zero noise, yet this discrepancy is discussed only briefly despite potentially affecting downstream conclusions. The manuscript should provide a more detailed analysis of why the recreated tournament differs from the original and whether this influences the subsequent noisy simulations. Likewise, the discussion relating the findings to historical nuclear false alarms is thought-provoking but should be presented explicitly as an illustrative analogy rather than empirical validation of the proposed framework, since real-world geopolitical decision-making involves numerous factors that extend well beyond the assumptions of the Iterated Prisoner's Dilemma.

Overall, this manuscript presents an interesting computational investigation and identifies a noteworthy empirical observation regarding the performance of classical IPD strategies under increasing implementation noise. However, the principal scientific claims currently exceed what is supported by the evidence. A more appropriate conclusion would be that adaptive, noise-tolerant strategies such as Downing become increasingly competitive under implementation noise, suggesting that robustness to isolated errors may unify several previously recognized strategic properties. Reframing the work in this manner, while providing a rigorous definition of flexibility, introducing quantitative measurements, expanding comparisons with prior literature and additional strategies, and strengthening the statistical analysis, would substantially improve both the scientific rigor and the credibility of the manuscript.

Reviewer 2

Rating scale questions

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The paper makes a significant contribution to scholarship.			✓		
The literature review was thorough given the objectives and content of the article.			✓		

Open response questions

Comments to author

Please verify that all links to the references point to the correct source.

The authors must disclose and acknowledge any assistance received in the preparation of this manuscript, including but not limited to editorial, technical, analytical, or writing support. All such contributions must be clearly stated in the Acknowledgments section.

Include enough recent references along with foundational ones. Ensure references directly support your claims. Avoid “padding.” Use credible sources (peer-reviewed journals, reputable books, official reports).

All assumptions (including implicit assumptions) must be explicitly stated and clearly justified. The authors should explain why each assumption is reasonable and discuss its impact on the results and conclusions.

Avoid overreaching conclusions that extend beyond what is supported by the data and analysis.

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